GREEN UNIVERSITY OF BANGLADESH

Department of Artificial Intelligence and Data Science
B.Sc. in Artificial Intelligence and Data Science (ADS)

Instagram Clone

Submitted By

Name: Md. Sani-Ul Islam
Student ID: 251035033
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Project Title

Instagram Clone Web Application

Abstract:

The Instagram Clone is a full-stack social media web application inspired by Instagram. The project was developed using Python, Flask, MySQL, HTML, CSS, and Bootstrap. It provides users with the ability to register accounts, securely log in, upload posts, interact with other users, and manage personal profiles. The purpose of this project is to demonstrate the practical implementation of software engineering concepts and database management systems in a real-world application.

Objectives:

The objectives of this project are:

- To develop a social media web application similar to Instagram.
- To implement secure user authentication.
- To manage relational data using MySQL.
- To demonstrate CRUD operations.

- To apply software engineering principles.
- To integrate backend and database technologies effectively.

Technologies Used

Technology	Purpose
Python	Programming Language
Flask	Web Framework
MySQL	Database Management System
SQLAlchemy	ORM for Database Operations
HTML5	Frontend Structure
CSS3	Styling
Bootstrap	Responsive User Interface
Jinja2	Template Engine
Werkzeug Security	Password Hashing
PyMySQL	MySQL Connector
Git & GitHub	Version Control

System Architecture

The application follows a three-tier architecture:

Presentation Layer

- HTML
- CSS
- Bootstrap
- Jinja Templates

Application Layer

- Flask Framework
- Routing
- Authentication
- Business Logic

Data Layer

- MySQL Database

- SQLAlchemy ORM

Features

User Registration

Users can create accounts by providing their information. Passwords are encrypted using hashing techniques for security.

User Login

Registered users can securely log into the system.

Profile Management

Users can manage and update their profile information.

Post Upload

Users can upload posts and share content.

Feed System

Users can view posts from the application feed.

Like and Comment System

Users can interact with posts by liking and commenting.

Follow System

Users can follow other users and build social connections.

Logout

Authenticated users can safely log out from the application.

Database Design

Main Tables

User Table

Contains:

- User ID
- Username
- Email
- Password (Hashed)
- Profile Information

Post Table

Contains:

- Post ID
- User ID
- Caption
- Image Path
- Upload Date

Comment Table

Contains:

- Comment ID
- User ID
- Post ID
- Comment Text

Like Table

Contains:

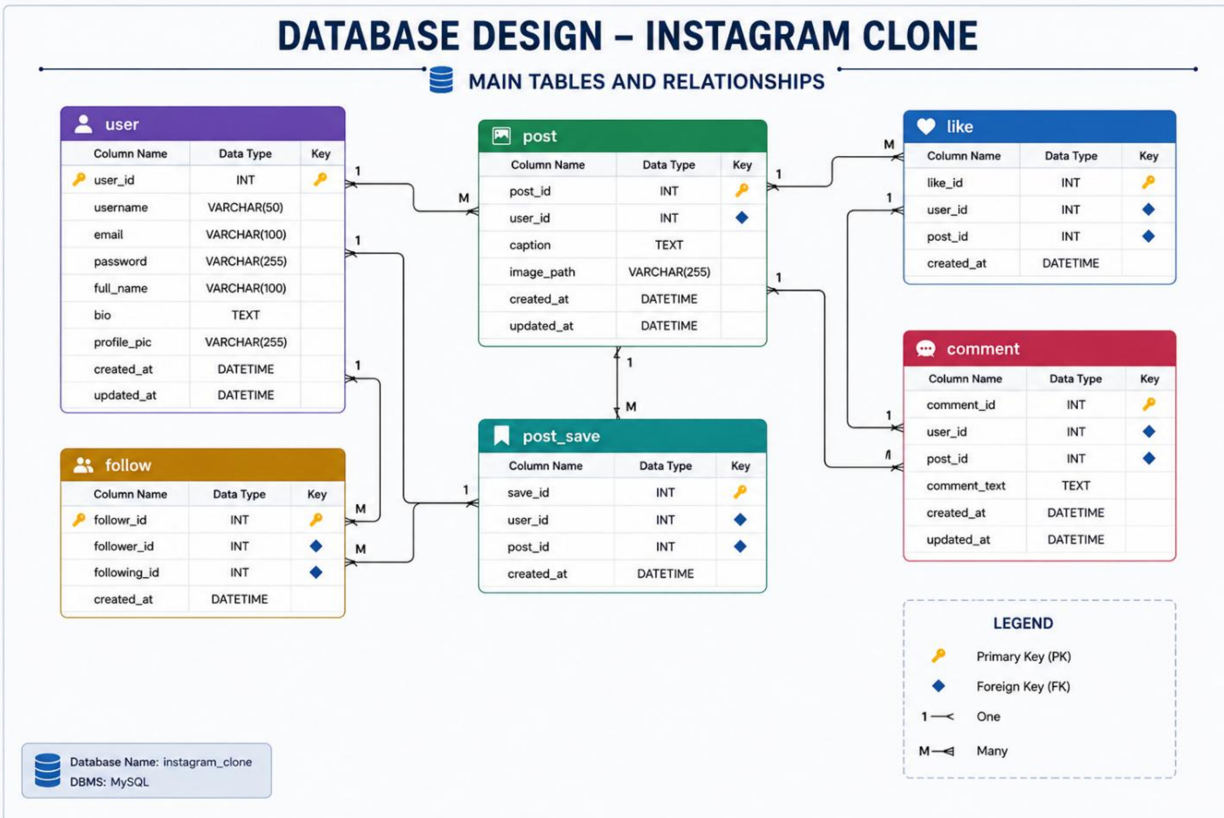
- Like ID
- User ID
- Post ID

Follow Table

Contains:

- Follower ID

- Following ID



Software Engineering Concepts Applied

- Object-Oriented Programming (OOP)
- MVC Architecture
- Modular Programming
- User Authentication
- Database Normalization
- Password Hashing
- Error Handling

Project Workflow

1. User Registration
2. User Authentication
3. Login Validation
4. Database Interaction

5. Post Creation
6. Feed Generation
7. User Interaction (Like, Comment, Follow)
8. Logout

Security Features

- Password hashing using Werkzeug Security.
- Protection against plain-text password storage.
- Database abstraction using SQLAlchemy.
- Form validation.
- Session management.

Advantages

- Easy to use interface.
- Secure authentication system.
- Efficient database management.
- Scalable architecture.
- Responsive design.

Limitations

- Real-time chat is not implemented.
- Story feature is unavailable.
- Push notifications are not included.
- Video upload support is limited.

Future Improvements

- Story Feature
- Real-Time Messaging
- Notifications
- Video Uploads
- Search Functionality

- Recommendation System
- AI-based Content Suggestions

Conclusion

The Instagram Clone project successfully demonstrates the integration of Software Engineering principles and Database Management concepts through the development of a full-stack social media application. The project provides practical experience in backend development, database design, user authentication, and web application architecture using Python, Flask, and MySQL. This project serves as a strong foundation for building larger and more advanced social networking platforms.

References

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